

Week 06

Assertion, Trigger and Stored Procedure

Cảm ơn thầy Trần Duy Quang đã cung cấp template cho môn học
Cảm ơn thầy Hồ Tuấn Thanh và thầy Đỗ Nguyên Kha đã xây dựng nội dung thực hành



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1

Notes

Put **ALL** SQL statements to **ONE** text (SQL) document names student ID 1651xxx_W06_yy.sql (yy is a self-grading mark 01-10):

- Comment Student ID and Full name on top of file
- Run ALL lines in the file without any error
- USE the RIGHT database

2

Use the **Company** database (week 4) with the following constraints:

- An employee's salary must not greater than her/his supervisor's salary (if any) and in the range [10000, 100000]
- A department must not have more than 10 employees

Function

Calculate age

```
CREATE FUNCTION fn_GetAge(@birthday date)
RETURNS int
AS
BEGIN
    RETURN YEAR(GETDATE()) - YEAR(@birthday)
END
```

Test function

```
SELECT dbo.fn_GetAge('1988-08-06')
```

Drop a function

```
DROP FUNCTION [ IF EXISTS ] dbo.fn_GetAge
```

Stored Procedure

Add a new employee

```
CREATE PROCEDURE sp_AddEmployee @id int, @code char(9), @firstName nvarchar(50),
@lastName nvarchar(50), @sex char, @departmentId int
AS
    IF EXISTS(SELECT COUNT(Id) FROM Employees WHERE DepartmentId = @departmentId
HAVING COUNT(Id) >= 10)
        THROW 50000, 'A department must not have more than 10 employees', 1
    ELSE
        INSERT INTO Employees (Id, Code, FirstName, LastName, Sex, DepartmentId)
VALUES (@id, @code, @firstName, @lastName, @sex, @departmentId);
GO
```

Execute a procedure

```
EXEC sp_AddEmployee 10, '123456789', 'Chi', 'Pu', 'F', 4
```

Drop a stored procedure

```
DROP PROCEDURE [ IF EXISTS ] dbo.sp_AddEmployee
```

Check constraint

Check salary with supervisor

```
ALTER TABLE Employees  
ADD CONSTRAINT ck_salary CHECK (Salary >= 10000 AND Salary <= 100000)
```

Trigger

```
CREATE TRIGGER tg_checkSalaryManager  
ON Employees  
FOR INSERT, UPDATE  
AS  
IF UPDATE(Salary)  
BEGIN  
    IF EXISTS(SELECT e.Id FROM Employees e JOIN Employees s ON e.SupervisorId =  
s.Id WHERE e.Salary > s.Salary)  
        THROW 50000, ' An employee's salary must not greater than her/his  
supervisor's salary (if any)', 1  
END
```

1. Lab

A: YY: 06

Database name: Company (Ref: Fundamentals of database systems)

Implement the following functions, stored procedures, and/or check/triggers for the following constraints

1. F1: Find the employee supervisor's full name (if not, return null)
2. F2: Get the total hours of an employee on all projects
3. S1: Add a new employee
4. S2: Add new project
5. C1: A department must not have more than 5 projects
6. C2: A department's manager must belong to that department

2. Homework

H: YY: 10

1. F3: Get the employee department's name
2. F4: Get the total hours of a department on project
3. S3: Add project
4. S4: Add hours to the work of an employee on a project
5. C3: Total hours of a project must not exceed 300 hours
6. C4: A project must not have more than 5 employees
7. C5: All employees in a project must belong to only 1 department
8. C6: An employee must not have more than 5 dependents
9. C7: Manager start date must not be after her/his birthday
10. C8: An employee must not supervise more than 5 employees